2026 9th National College Student Embedded Chip & System Design Competition

基于FPGA的细分编码器

A 2,048-ppr magnetic encoder on one Artix-7: 16-bit on-chip interpolation, closed-form self-calibration, ABZ / UVW / SPI / UDP

FPGA Innovation Track · AMD Track · Independent Topic (Junior Group) · XC7A35T-FGG484-2

THE PROBLEM

Optical encoders **lose accuracy** in dust, oil and vibration; magnetic sensing is rugged, but the raw sin/cos bridge is never ideal.

Imported interpolation ASICs **fix both the algorithm and the interface**: multi-turn counting needs a second chip, and the calibration recipe stays opaque.

Goal: **2,048 ppr, 16-bit angle, sub-µs latency**, with ABZ, UVW, SPI and Ethernet from one 50 MHz clock domain — no MMCM, no soft-core.

OUR APPROACH

Dual 12-bit AD9238 samples the SMR bridge at 65 MSPS; synchronous capture and ÷4 decimation give 12.5 MSPS.

Six-parameter closed-form calibration — sin/cos offset, gain ratio, quadrature phase, residual shift — no LUT, no host tuning.

17-stage CORDIC atan2 and a type-II PI loop yield 16-bit angle and speed; incremental unwrapping keeps direction across 0/2π.

SIGNAL PATH — ONE CLOCK DOMAIN, 41.5-CYCLE LATENCY (≈830 NS, DESIGN)

Front end
SMR bridge · AD9238
12-bit ×2 @ 65 MSPS

→

Capture ÷4
synchronous sampling
12.5 MSPS clean data

→

Calibrate + atan2
6-parameter correction
17-stage CORDIC, 16-bit

→

Track + output
type-II PI tracking
speed, direction, unwrap

shared 16-bit angle + speed + status → output stage

ABZ
1…1024× interpolation
65,536 edges/rev at 32×

UVW + Z
six-step commutation
Z = 12.00 Hz at 720 rpm

SPI slave
32-bit angle word
register map, 2 Mbaud

UDP / 1 GbE
28,672-point captures
9,968 fps streaming

On-chip TFT UI
1024×600 touch
5 diagnostic pages

BOARD-VERIFIED RESULTS — V74 PRODUCTION BITSTREAM

2,048
pulses / rev
2,048 distinct counts measured across a full turn

25.27
kHz electrical
sin/cos at 720 rpm;
≈495 samples per cycle

719–731
rpm
12.0–12.2 Hz mechanical,
Z pulse at 12.00 Hz

598,115
frames / 60 s
9,968 fps Ethernet soak,
radius drift < 0.02 %

Timing, 50 MHz	WNS +0.409 ns · WHS +0.025 ns · 0 failing endpoints
Logic	10,926 LUT (52.53 %) · 8,697 FF (20.91 %) · 17 DSP
Memory	32 / 50 BRAM · single clock · DRC 0 errors
Bitstream	1,512,664 B · SHA E4634907…DA2680 · build 0x0018
Latency	41.5 cycles ≈ 830 ns, sample to angle (design budget)

How it was proven

Two silent deadlocks on hardware: sign-extended phase at 32,768, and a swallowed reverse wrap at sector 0/5.

Build-ID bisect: probe 0x0013 traced 10 s of silence to one step of −65,403 instead of +133.

tb_acc2 reproduces 0 → 6 → 12 counts for 3,000 × +133 steps.

WHAT IS NEW

Closed-form 6-parameter calibration — one pass, no lookup table, no host iteration.

Wrap-safe incremental unwrapping — direction and multi-turn count survive every 0/2π crossing.

Symmetric forward/reverse UVW — a wide-base branch makes the 0→5 sector edge identical both ways.

Single clock, no MMCM, no soft-core — 41.5 cycles sample-to-angle, deterministic.

Self-hosted diagnostics — 1024×600 touch UI, 1 GbE capture, ILA and register map.

REPRODUCIBILITY & EVIDENCE

MIT-licensed repository: RTL, constraints, host tools, testbenches and the full 59-page report.

263 evidence files, 283 SHA-256 lines — every number here traces to a report, log or bitstream.

54 simulation benches plus one-command sign-off: final_verify.ps1 checks sources, timing, bitstream SHAs, documents.

LLM-collaboration log and skills package recording how the two deadlocks were found and fixed.